

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1

Year: 2017

Subject: 392133 Physics III

Section: 15-16

Date: 5 October 2017

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. The examination has 7 pages (including this page), 6 problems and a total score of 60 points.
2. Write all your solution and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are NOT allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious
offence which will result in expulsion from the University.

1.1 [5 points] A block attached to a spring (the spring constant=100 N/m) oscillates in simple harmonic motion along the y axis. The limits of its motion are $y = 0$ cm and $y = 20$ cm and its period of the oscillation is 0.5 s. Determine its mass, amplitude of oscillation and frequency.

1.2 [5 points] A 3 kg block, attached to a spring, executes simple harmonic motion according to $x = 2\cos(50t)$ where x is in meters and t is in seconds. Find the spring constant of the spring and maximum velocity of the block in m/s.

2.1 [5 points] A particle is in simple harmonic motion along the x axis. The amplitude of the motion is x_m . When it is at $x = x_1$, its kinetic energy is $K = 5$ J and its potential energy (measured with $U = 0$ at $x = 0$) is $U = 3$ J. When it is at $x = -x_m$, determine the kinetic and potential energies.

2.2 [5 points] The period of a simple pendulum is 1 s on Earth. When brought to a planet where g is one-tenth that on Earth, find its period.

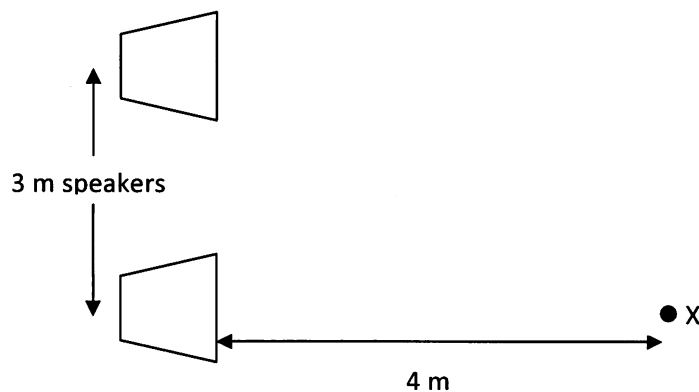
3.1. **[5 points]** Sinusoidal water waves are generated in a large ripple tank. The waves travel at 20 cm/s and their adjacent crests are 5.0 cm apart. Determine the time (period) required for each new whole cycle to be generated.

3.2 **[5 points]** If the length of a piano wire (of given density) is increased by 5%, what approximate change in tension is necessary to keep its fundamental frequency unchanged?

4.1 [5 points] In our experiment, when a certain string is clamped at both ends, a student reports that the lowest four resonant frequencies are measured to be 100, 150, 200, and 250 Hz. However, we know that one of the resonant frequencies (below 200 Hz) is missing. What is it?

4.2 [5 points] If the sound level is increased by 10 dB, by what factor has the intensity of the sound increased?

5.1 [5 points] Two small identical speakers are connected (in phase) to the same source. The speakers are 3 m apart and at ear level. An observer stands at X, 4 m in front of one speaker as shown. If the amplitudes are not changed, find the wavelength of the sound that he hears at the most intense.



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5.2 [5 points] An organ pipe with both ends open is 0.85 m long. Assuming that the speed of sound is 340 m/s, find the frequency of the third harmonic of this pipe.

6. [10 points] A source emits sound with a frequency of 1000 Hz. Both it and an observer are moving in the same direction with the same speed, 100 m/s. If the speed of sound is 340 m/s, what is the frequency of the sound that the observer will hear?

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